

Official statement (Cell 3). Any $d + 1$ lines in $\mathbb{R}^d$ satisfy $S \leq \left(\binom{d+1}{2} - 1\right)\pi/2$, i.e. $A \geq \pi/2$.

Here S is the sum of the acute angles between all pairs of lines. For unit representatives $x_1, \dots, x_N$, write

$$A = \sum_{i < j} \arcsin |\langle x_i, x_j \rangle|.$$

Since $S = \binom{N}{2}\pi/2 - A$, the official statement is equivalent to $A \geq \pi/2$ when $N = d + 1$.

Proposition 1 (Orthogonality saturation). *For fixed N and $d \geq 2$, there is a maximizer of S such that each line is either orthogonal to every other line, or the lines orthogonal to it span a subspace of dimension at least $d - 1$.*

Proof. The space of N -tuples of lines is a product of compact projective spheres, and S is continuous. Choose a maximizer with as many orthogonal pairs as possible. Fix its line ℓ_i and let W be the span of the representatives of all lines orthogonal to ℓ_i . Suppose $\dim W \leq d - 2$ and ℓ_i is not orthogonal to every other line. A unit representative x_i lies in $W^\perp$, whose dimension is at least two. Choose a two-dimensional plane $U \subset W^\perp$ containing x_i such that the inner product with one nonorthogonal line varies along its unit circle. Such a plane exists: the projection of that line onto $W^\perp$ is nonzero, and one can choose the other direction of U accordingly. Rotate only x_i in U .

Consider the closed interval between the two consecutive positions where a new orthogonality occurs, one on either side of the original position. There are such positions because the varying nonzero inner product is sinusoidal and has zeros. Throughout the open interval, every previously nonorthogonal inner product has fixed sign. After a possible sign change, its corresponding angle term has the form

$$f(t) = \arccos(r \cos(t - \phi)), \quad 0 \leq r \leq 1, \quad \cos(t - \phi) > 0.$$

For $0 < r < 1$, direct differentiation gives

$$f''(t) = \frac{r(1 - r^2) \cos(t - \phi)}{(1 - r^2 \cos^2(t - \phi))^{3/2}} \geq 0.$$

For $r = 0$ the term is constant; for $r = 1$ it is affine away from coincidence, with a convex corner at coincidence. Thus the sum of all varying angle terms is convex on this interval, including any coincidence points. At least one endpoint has value at least the value at the original position. Global maximality makes that endpoint another maximizer. Every old orthogonality involving ℓ_i persists because the rotation stays in $W^\perp$, while an additional orthogonal pair appears. This contradicts our choice of maximizer. Hence the stated alternative holds for every line. $\square$

Proposition 2 (Cell 3). *For every $d \geq 1$, any $d + 1$ lines in $\mathbb{R}^d$ satisfy*

$$A \geq \frac{\pi}{2}, \quad S \leq \left(\binom{d+1}{2} - 1 \right) \frac{\pi}{2}.$$

Proof. For $d = 1$, the two lines coincide, so $A = \pi/2$. Suppose $d \geq 2$. It is enough to prove the inequality at a maximum of S ; choose the saturated maximizer furnished by the preceding proposition. Let J join two indices exactly when their lines are *not* orthogonal. A vertex adjacent in J to nobody is harmless. At every other vertex, the orthogonality neighbors span dimension at least $d - 1$, so there are at least $d - 1$ of them among the other d vertices. Consequently J has maximum degree one: it is a matching, together with isolated vertices.

Distinct connected components of J are mutually orthogonal. Each component consists of one line or a pair of nonorthogonal lines; it has rank one or two, with a pair having rank one precisely when the two lines coincide. If every component had rank equal to its number of lines, their mutually orthogonal spans would have total rank $d + 1$, impossible in $\mathbb{R}^d$. Therefore one paired component has rank one. Its two coincident lines contribute $\arcsin 1 = \pi/2$ to A . All other contributions are nonnegative, giving the claim. $\square$

Sharpness and equality at a saturated maximizer. Take an orthonormal basis of $\mathbb{R}^d$ and repeat one basis line. There is exactly one nonorthogonal pair, which contributes $\pi/2$; thus equality holds. Conversely, the matching argument shows that every *saturated maximizer* with equality has exactly this form, up to an orthogonal transformation and relabeling: its required coincident pair exhausts the A -budget, so every other pair is orthogonal. This does not assert that arbitrary equality configurations have this form; for instance, equality for three distinct lines in a plane is also possible.